

Room 5 (P5 W_T non-D-finite): moderator summary

2026-09-26

Two seats (Hilbert, Erdős), each attempting a route different from the already-tried catalytic-variable kernel method (P5_v2_note.tex).

Hilbert: NOTHING WORKED, three routes precisely blocked.

A contour/Cauchy-integral kernel gives only a tautological restatement, since the reducibility relation (division by $2\mu_T$) has an infinite, non-compact index set, unlike the finite-state travel block behind U, V, G_0^T . A special-function (theta/ q -Bessel) substitution fails: ζ_T is fixed, not a deforming/confluent parameter, and the carry's length weight is a min-plus optimum, not a closed quadratic form. A rationality transfer via the quasi-isometry to $DL(5, 5)$ fails because that rationality result is for finite-lamp lamplighter groups, while H_T 's lamp group A_T has rank 2 and is not finitely generated – the QI does not preserve the exact generating function.

Erdős: PARTIAL, same conclusion via different attempts. A two-integer (real/imaginary-coordinate) re-encoding of the catalytic step, via the monic recursion for $\eta = c\zeta$, relocates rather than removes the obstruction: the integer-pair recursion itself grows geometrically ($\sim c^h$) per step, reproducing the same unbounded-carry fact (independently reconfirming, in a different basis, that A_T is infinite – consistent with, not beyond, the already-known result). A tower of finite-radius truncations $F_T^{(N)}(x)$ (each individually rational/D-finite, converging coefficientwise to F_T) is a genuinely new formal object – paper1 only states the analogy to Akiyama–Frougny–Sakarovitch in words – but provides no new leverage; it repackages the unproved carry-width growth law rather than resolving it.

Room verdict: NO PROGRESS, but with two independently-reached, precisely-diagnosed structural obstructions (rank-2 non-f.g. lamp group defeats the QI-rationality shortcut; the min-plus/non-contracting-base structure defeats every continuation route tried) and one small, honest formal artifact (the truncation tower) that clarifies the picture without advancing

it. P5 stands exactly where `P5_note.tex`/`P5_v2_note.tex` left it: OPEN, with Gaps 1–3 now attacked from two more angles, both closed off for stated reasons.